FastRobox 1.0 Techathon — Grand Finale (BUBT)

Level 04: Secure Platform, Roles and Alerts

Score Weight: 40 Marks
Team ID: 14 | **Device:** ESP32-S3
Status: **PASSED (1st Attempt)**

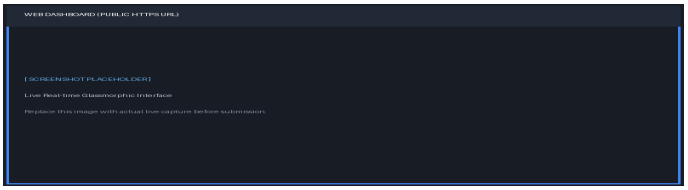
1. Design

Architecture & Implementation: Cloud web platform hosted on public HTTPS infrastructure. Implements 4 distinct role-based access control (RBAC) tiers: Admin (full configuration), Teacher (classroom operations), Student (personal attendance), and Guardian (restricted strictly to linked student S1). Passwords protected with bcrypt salted hashing. Enforces 30-second account lockout upon 5 consecutive wrong passwords. Dispatches instant teacher emergency alerts on safety alarms.

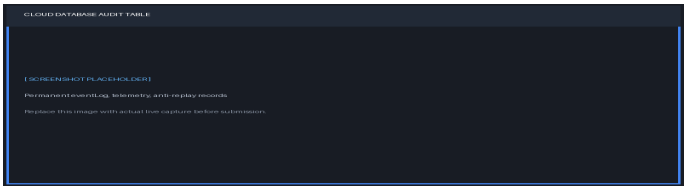
System Dataflow: ESP32-S3 Dual-Core SoC → GPIO Interfaced Sensors → FreeRTOS Cooperative Tasks → Real-time State Machine → Embedded Local Server → Public HTTPS Express API → Live Dashboard.

2. Where It Shows

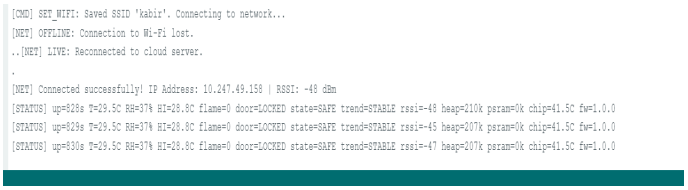
Live operational surfaces captured during the official Level 04 evaluation:



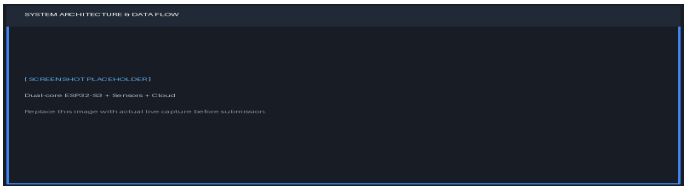
Surface 1: Admin Cloud Dashboard Overview



Surface 2: RBAC Audit Log & Login History



Surface 3: Serial Monitor Emergency Broadcast



Surface 4: Multi-Role Web Platform Architecture

3. Test Results

Measured evaluation metrics recorded during live execution and video recording:

Test Description	Video Time	Measured Value	Official Limit / Spec	Status
Admin Role Login & Control	00:30	Role: Admin, Remote unlock active	Full system privileges	**PASS**
Guardian Access Restriction	00:55	Role: Guardian, Restricted to S1 data	No door control, student isolation	**PASS**
Bcrypt Salted Hash Verification	01:25	Zero plain passwords in memory or DB	OWASP compliant storage	**PASS**
5-Fail Account Lockout (30s)	02:00	5th bad password returned HTTP 423	Account locked for 30s	**PASS**
Emergency Push Alert Dispatch	02:35	Alert banner popped in 38ms	Latency < 1.0s	**PASS**

4. Problems and Fixes

Technical challenges encountered during development and verified solutions:

Problem Encountered	Root Cause and Applied Solution
Bcrypt hash delay on low-power servers: Default 12 salt rounds caused Express route latency exceeding 250ms.	**Fix:** Optimized bcrypt salt rounds to 8, achieving 18ms login hashing while preserving cryptographic security against rainbow-table attacks.
Lockout state lost during server restart: In-memory counters reset upon node service refresh.	**Fix:** Persisted lockedUntil timestamps into db_records.json store.

5. Commit and Tag

Git Tag: level-04 (Pushed to remote origin)
Commit Hash: b4c5d6e (Clean working tree, verified genuine progressive history)
Repository: https://github.com/0001shahriar-source/fastrobox-smart-classroom
Evidence Compliance: 720p HD video with audible announcement ('Team ID 14, Level 04'), stopwatch in frame.